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2.4.10 Octal to Binary Conversion

For each Octal digit in the given number write its 3 digits binary equivalent using positional notation.

Example

Convert $(6213)_8$ to equivalent Binary number

6	2	1	3
↓	↓	↓	↓
110	010	001	011

$(6213)_8 = (110010001011)_2$

Workshop

5. Convert the following Octal numbers into Binary numbers.
(A) 472 (B) 145 (C) 347
(D) 6247 (E) 645

2.4.11 Hexadecimal to Decimal Conversion

To convert Hexadecimal to Decimal we can use positional notation method.

1. Write down the Hexadecimal digits and list the powers of 16 from right to left(Positional Notation)
2. For each positional notation written for the digit, now write the equivalent weight.
3. Multiply each digit with its corresponding weight
4. Add all the values to get one final value.

Example

Convert $(25F)_{16}$ into its equivalent Decimal number.

Weight	256	16	1
Positional Notation	16^2	16^1	16^0
Given number	2	5	F(15)

$(25F)_{16} = 2\times256 + 5\times16 + 15\times1$
$= 512 + 80 + 15$
$(25F)_{16} = (607)_{10}$

2.4.12 Hexadecimal to Binary Conversion

Write 4 bits Binary equivalent for each Hexadecimal digit for the given number using positional notation method.

Example

Convert $(8BC)_{16}$ into equivalent Binary number

8	B	C
↓	↓	↓
1000	1011	1100

$(8BC)_{16} = (100010111100)_2$